

Pre-Lecture Assignment 3

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1 Problem 1)

We can divide DPG equation

$$\nabla_{\theta} V^{\pi_{\theta}}(\mu) = \frac{1}{1-\gamma} E_{s \sim d_{\mu}^{\pi_{\theta}}} [\nabla_{\theta} \pi_{\theta}(s) \nabla_a Q^{\pi_{\theta}}(s, a)|_{a=\pi_{\theta}(s)}]$$

into three parts:

1. $E_{s \sim d_{\mu}^{\pi_{\theta}}}$: The state s should follow the π_{θ} policy and the initial state should follow the distribution of μ .

2. $\nabla_{\theta} \pi_{\theta}(s)$: This term represents how the policy parameters θ affect the chosen action a in a given state s .

3. $\nabla_a Q^{\pi_{\theta}}(s, a)|_{a=\pi_{\theta}(s)}$: This represents the gradient of the Action-Value function with respect to the action. It indicates the direction in which the action should be changed to most rapidly increase expected return.

The difference between DPG and Stochastic PG, For stochastic PG, it integrates over both state and action spaces. It requires sampling actions from distribution, which can lead to high variance in the gradient estimates, especially in high-dimensional action spaces. As for Deterministic PG (DPG), it only integrates over the state space. Because the policy is deterministic, we don't need to sample multiple actions per state to estimate the gradient. The results in a more computationally efficient update with significantly lower variance.

2 Problem 2)

Off-Policy learning concept: Off-policy learning is a reinforcement strategy where the target policy (the

policy being learned and optimized) is different from behavior policy (the policy used to interact with the environment and generate data).

Off-policy example: Training a robotic arm to perform a precision task using data collected from several different robots or even human demonstrations. The reason why it uses Off-policy: The robot being trained (Target Policy) learns from a Replay Buffer containing experiences generated by other agents or previous versions of itself (Behavior policy). The data used for learning doesn't necessarily reflect the current robot's decision-making logic, allowing it to learn from a diverse set of historical successes and failures without needing to perform those exact actions again.

3 Problem 3)

DPG is particularly suitable for continuous action spaces for the following reasons:

1. Elimination of Action Integration: In continuous spaces, finding the optimal action for a stochastic policy requires complex integration over an infinite number of possible actions. But DPG can avoid this by simply following the gradient of the Q-function.

2. Efficiency in High Dimensions; For robots with many degree freedom, the action space is vast. DPG provides a direct path to optimize the action by treating it as a continuous optimization problem, which is much faster than the trial-and-error sampling required by stochastic policies.